

Response to Reviewers

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When does sampling uncertainty matter in matrix population models? Evidence from published projection matrices

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We thank the editors and both reviewers for their careful and constructive assessment of our manuscript. We were very pleased that the reviewers found the work to be original and valuable, and we found their comments genuinely helpful in strengthening the paper.

We have made substantial revisions in response, and we respond to every point individually below. The reviewers' comments are shown in grey italics followed by our responses. Line numbers refer to the revised, line-numbered manuscript.

To summarise the main substantive changes: These are: (1) a new simulation study with known truth as Analysis 3, replacing the previous multi-site *Astragalus* analysis. We move that analysis to become a supplementary robustness check on Analysis 2, (2) a new analysis quantifying how the displacement of point estimates from their sampling distributions scales with boundary-survival structure, (3) an empirical comparison of included versus excluded studies, and a characterisation of where boundary estimates and small sample sizes occur, (4) 95% intervals added to Table 1, (5) a sensitivity analysis for the quasi-stable-distribution truncation threshold, and (6) an improved consistent colour scheme for the figures.

Associate Editor

Both reviewers and I recognize the importance and originality of your work. This is an article of great relevance to comparative demographic studies, which are perhaps the most influential applications of population projection models. [...] To my knowledge, this manuscript is the first to quantify the effect of estimate uncertainty on demographic metrics and, as a result, on comparative demographic analyses.

Response: We thank the Associate Editor for this positive assessment. We address the specific points raised by the Associate Editor and by both reviewers in detail below.

Both R1 and R2 found that you should characterize the bias of your dataset in a better and more systematic way, and you should characterize its effects on your inferences. In particular, they indicated to examine taxonomic bias, and bias in the type of data that you selected (e.g. which species are we more likely to find boundary survival estimates? Which species tend to have smaller sample sizes?).

Response: We agree and have addressed this in full. See our responses to Reviewer 1, Major Comment 1 (included-versus-excluded species) and to the taxonomic-representation and boundary/small-N characterisation points raised by both reviewers, where we describe the new Table S11, Table S12, and associated text.

R1 has also a set of important suggestions. First, regarding boundary survival estimates, R1 suggests to perform an analysis to explain and quantify the discrepancy between life expectancy/shape and their sampling distribution. Second, regarding Table 1 and its variance comparison, R1 suggests to prominently discuss the implications of ignoring temporal variation. [...] Third, in analysis 2, R1 suggests to include a version of the point estimate model using a Poisson likelihood.

Response: These correspond to Reviewer 1's Major Comments 2, 3, and 4, to which we respond in detail below (a new displacement analysis, more prominent treatment of the temporal-variation caveat, and a new simulation study).

R2 also criticized your justification and choice of analyses. First, R2 suggests to better explain and justify the relevance of analysis 1. Second, R2 found that analysis 3 mostly replicated analysis 2. Instead, R2 suggests performing a simulation analysis (exploiting known climate coefficients) instead.

Response: We have strengthened the motivation for Analysis 1 and, as recommended, replaced the former Analysis 3 with a simulation study using known climate coefficients (the “known truth”).

Associate Editor — minor comments

260-262: Unclear to the reader how these three options differ. What's the difference b/w pooled and mean MPM? And how do these differ from a "population-level matrix"?

Response: We have revised this sentence to make the three matrix types unambiguous. The text now distinguishes a pooled population MPM (a matrix constructed by pooling individual-level demographic information across periods and/or populations), a population mean MPM (an element-by-element arithmetic mean across annual matrices), and our fallback case, where we constructed a population-level matrix as the element-by-element mean of the available annual 'Individual' matrices in COMPADRE. [Lines 291-298]

351-353: This section is confusing. The equations are outside of the sentences (e.g. Eq. 11 starts after a full stop, and there is no full stop after Eq. 12), so it's hard to understand the context here.

Response: We have corrected this: the equations are now embedded within the surrounding sentences with appropriate punctuation.

Reviewer 1

Major comments

1. Sample selection bias is insufficiently addressed. [...] I would encourage the authors to strengthen this point with an empirical comparison of key characteristics between included and excluded studies (e.g., matrix dimension, life form, mean survival rates, proportion of boundary estimates).

Response: We agree and have now added an empirical comparison of the studies retained in the analysis, versus those not retained. Specifically, we compared study-level mean matrix dimension, dominant life form, mean stage-specific survival, and the proportion of boundary survival estimates (0 or 1) under the current reproducible COMPADRE-based screen (Table S11 and Supplementary Methods S1.8). These analyses show that included and excluded studies are broadly similar in matrix dimension, life form, and mean survival, but that included studies have a higher prevalence of boundary survival estimates, especially survival estimates of exactly 1. We have replaced the previous unsupported statement (“no strong reason to expect systematic differences”) with this summary in the Methods. [Lines 143-148]

2. The boundary estimate problem is central but underexplored mechanistically. [...] a regression or even a stratified or binned comparison showing how the difference between point estimates and sampling distribution estimates scales with the number of boundary stages or with the proportion of individuals in boundary survival stages would help clarify the mechanism.

Response: We have added an explicit analysis of displacement magnitude in Analysis 1. For each population-level matrix, we quantified displacement as the absolute difference between the point estimate and the median of the sampling distribution (for mature life expectancy, on the log₁₀

scale), and related this to (i) the number of boundary survival stages and (ii) the proportion of sampled individuals in boundary-survival stages. Both predictors were positively associated with displacement magnitude for both shape and life expectancy, consistent with the proposed mechanism but not establishing causality. These results are reported in new Supplementary Methods and Results subsections, with a new figure and table (Figure S3 and Table S13). [Lines 561-563]

3. The variance-component framing is potentially misleading. [...] pooling sample sizes across years may underestimate true sampling uncertainty [...] the reported ratios [...] are likely conservative. [...] the extent to which variation currently attributed to "biological variation" in comparative analyses of ageing may instead reflect a combination of sampling noise and temporal process variation [...].

Response: We have corrected the Table 1 description: its ratios compare across-entry variances of matrix point estimates and reconstructed posterior means. They do not necessarily show how much of the variation is due to sampling noise rather than genuine biological differences and because stage sample sizes were pooled across years, the posterior distributions may actually underrepresent temporal variation. We now discuss the possible implication for comparative senescence. [Lines 333-338, 443-446]

4. Analysis 2 conflates structural model change with uncertainty propagation. [...] including a version of the point-estimate model using a Poisson likelihood (without uncertainty propagation) as an additional comparison would help isolate the effect of uncertainty from that of the structural model differences.

Response: We agree that our original Analysis 2 confounded the change in likelihood structure (Gaussian on log per-capita recruitment versus Poisson on raw counts with an offset) with the propagation of sampling uncertainty. Rather than adding a separate Poisson point-estimate model within the *Silene* analysis, we explored related model differences through the new simulation study (Analysis 3). That analysis fits all three formulations to data with known truth, with a log link for the Gaussian comparator so that all three coefficients target the same log-scale climate effect. Both Poisson models account for count sampling variation. [Lines 381-385 and Figure 6]

5. Generalisability claims exceed the scope of the data. [...] the proposed thresholds [...] should be interpreted as heuristics derived from a restricted and taxonomically narrow dataset rather than formally validated decision rules. Similarly, the conclusion that integrated metrics such as λ are "often robust" would benefit from additional qualification.

Response: We agree - this is an important point. We now present these thresholds as provisional screening heuristics informed by our restricted dataset, not as super-robust validated decision rules. We also limit our claims of lower sensitivity for integrated metrics to the dataset analysed and explicitly state that the proposed favourable conditions do not guarantee robustness. [Lines 565-567, 587-594]

Minor comments

1. Table 1 needs confidence intervals. [...]

Response: We have added 95% intervals to Table 1 by resampling population-level entries in pairs and recalculating the point-estimate-to-posterior-mean variance ratio for each bootstrap sample. These intervals describe uncertainty in that ratio.

2. The shape metric is not self-explanatory. [...] clarifying what a shape of +0.3 versus -0.2 implies for how a plant ages [...].

Response: We have added a short sentence in the Methods section, just after the definition of Type I/II/III survivorship curves, clarifying the biological meaning of the metric. We say, “Thus, more positive values indicate that mortality is concentrated later in adult life, whereas more negative values indicate that mortality is concentrated earlier in adult life.” [Lines 282-284]

3. The quasi-stable distribution truncation rule is consequential but briefly justified. [...] readers deserve some sensitivity analysis showing how results change if a different threshold (e.g., 5% or 0.5%) is used.

Response: We added now a sensitivity analysis for the quasi-stable-distribution truncation threshold used in the shape calculation. Recomputing shape at 0.5% and 5% thresholds showed very high concordance with the 1% threshold used in the main analysis, with only small median shifts in shape. We report the check in Supplementary Methods S1.13, Supplementary Results S2.4, Table S14, and in Figure S4.

Reviewer 2

Overview: [...] the work includes an impressive set of analyses and code that could be extremely useful [...]. However, there are a few points where the manuscript needs substantial edits [...]. Primarily, the motivations for the sets of analyses should be more clearly presented [...] and the results could be grounded in a more biologically interesting way [...].

Response: We thank Reviewer 2 for this constructive overview and for the encouraging comments on the analyses and code. We have sharpened the motivation for the set of analyses (including a clearer statement of the ‘spectrum’ of types of biological analysis they span). We also replaced the former Analysis 3 with a simulation study, and grounded the results more explicitly in their biological implications, as we explain further below.

Abstract

Line 8: In the abstract, as in the introduction, it would have been valuable to more clearly define the biological questions that motivate the three analyses.

Response: We have now revised the abstract to define the three analyses explicitly: i.e a multi-species comparative analysis of the life expectancy-shape relationship, a single-species time-series analysis of climate effects on recruitment, and (now) a simulation study with known truth. [Lines 11-14]

Line 14: It is not clear what “it” refers to in the second sentence here because “trait” is not very clearly defined.

Response: We have removed the ambiguous “it” so the sentence now reads “Accounting for sampling uncertainty barely shifted integrated metrics such as population growth rate” [Lines 15-18]

Line 26: The overall goal of the paper is really important and appreciated!

Response: Thanks!

Introduction

Line 40: [...] I would suggest re-organizing the introduction to start with the broader idea of sampling uncertainty, and then point out that it is particularly relevant to these sorts of analyses that re-use published matrices where the uncertainty is even more unavailable.

Response: We have reorganise the start of the Introduction to begin by talking through broad relevance of sampling uncertainty across demographic analyses. This takes the reader from raw-data and single-species studies up to large comparative studies, before narrowing to the compiled-database setting where the uncertainty information is least available. [Lines 35-48]

Line 55: [...] is the trait for a population or for a species, and is it different from a life history trait[?]

Response: We have clarified this terminology at first use in the Introduction. The text now defines derived demographic traits as population- or species-level quantities computed from MPMs, and distinguishes life-history traits as one subset of those quantities. [Lines 63-68]

Line 65: [...] I do think it would be useful for this manuscript to more clearly present the methods by which the authors are incorporating this uncertainty.

Response: We have now made the uncertainty-propagation methods we use more prominent. The 'Sampling distribution for a single MPM' section now describes direct conjugate posterior sampling, including the priors, draw counts and treatment of zero entries and unavailable sample sizes, and each analysis states which quantity carries the propagated uncertainty and how it enters into the statistical model. [Lines 179-204]

Line 78: I suggest phrasing this more as "deriving parameters from MPMs that have uncertainty, and then using those parameters to test for statistical relationships" [...].

Response: We have rephrased this passage to make the chain of propagation explicit. Going from uncertainty in the sampled transition rates, through the derived parameters, to the statistical relationships tested, rather than referring more generically to incorporating estimates into a model. [Lines 103-104]

Line 87: How these analyses bracket this "spectrum" was not super clear to me, and [...] there was basically no introduction of the analysis of climate drivers. [...] A paragraph to introduce climate-driven demographic forecasting or how species responses to climate differ would be really helpful [...] and the difference between implicit year-to-year variation in matrices vs more explicit climate driver variation.

Response: We now clarified the roles of the three analyses in the Introduction: a comparative analysis of lifetime-integrating traits, a within-species climate-demography analysis, and a simulation assessing bias and interval coverage against known climate effects. The multisite *Astragalus* analysis is now "downgraded" to a supplementary robustness check rather than removing entirely. A dedicated paragraph outlines the motivation for climate-demography analyses and explains that independent, additive, unbiased response error in linear regression reduces precision rather than attenuating the slope. Its consequences in demographic models depend on the error structure and model formulation. [Lines 78-88, 105-116]

Line 95: The goal of having diagnoses seems like a very useful contribution, but what these metrics are was not clear to me after having read the full manuscript. Can there be more of a summary of this material as a table or box, somewhere in the manuscript?

Response: We have added text to the introduction that we also provide a practical diagnostic checklist for COMPADRE users, and the checklist itself is set out in the Discussion. [Lines 120-122]

Methods

Line 122: It was not clear to me if all studies had at least 4 years of temporal replication or not?

Response: We have now clarified the matrix model selection criteria. The 'at least 4 annual matrices' requirement applied only to the 5 pre-2010 studies added to support the time-series analyses... it was not a selection criterion for the full 31-study dataset used across all analyses. [Target studies section, Lines 138-140]

Line 147: [...] a statistically estimated model with many more stages or ages, or even IPMs, are these likely to have less drastic consequences of uncertainty because each individual bin is sharing information across stages?

Response: We have added a short paragraph to the Methods section to clarify that uncertainty may differ when transition rates are estimated with other types of model, such as information-sharing models like hierarchical vital-rate models or IPMs. Partial pooling will often reduce the variance of weakly informed transition estimates, but may also induce stronger correlations across transitions. In these cases we think uncertainty should ideally be propagated from the joint posterior of the fitted vital-rate model rather than treating transition estimates as independent. [Lines 169-178]

Line 168: The choice of Poisson distribution for bootstrapping reproduction is probably reasonable, but I wonder whether that fully captures the uncertainty [...]. This also makes me wonder if there are other reproduction-centric population metrics [that] would have different patterns compared to [...] survival, population growth and life expectancy?

Response: We now have acknowledged the overdispersion point in the Methods: a Poisson distribution for fecundity may under-represent uncertainty when reproduction is overdispersed. We agree that reproduction-focused quantities, such as net reproductive rate, may show different sensitivities to fecundity uncertainty. We have not tested those quantities, so their robustness cannot be inferred from our recruitment results. [Lines 195-196]

Line 175: It would be valuable to provide more sample sizes for the numbers of matrices for which there was no useable sample size.

Response: We have now reported this explicitly: “For 5 of the population matrices (one population each), we could not recover even the per-stage sample sizes, so we had no basis for estimating their uncertainty. Where these matrices were kept, the affected columns were held at their published values rather than given reconstructed sampling uncertainty.” [Lines 306-311]

Line 185: Is it the case that the sample size, and the total estimate population size is available for each of the included studies?

Response: We have clarified that our analyses use recovered stage-specific or transition-specific sample sizes, not total estimated population size, which was not systematically available across the included studies and was not used in the analyses. [Lines 306-308]

Line 190: It is not clear how the extra row of death frequencies is used.

Response: We agree it was unclear, and have clarified it in the Methods. The revised text explains that the extra row (d) represents death, the remaining outcome for individuals that did not survive, and states that, in the posterior sampling, death probability is drawn jointly with the living-transition probabilities in each U-column, but only the living-stage components are written back into the simulated U matrix. [Lines 218-221]

Line 214: This paragraph is somewhat repetitive of paragraph at line 80 [...]. [...] it would still be useful to more clearly introduce why readers would care about climate-demography relationships in their own right.

Response: We have tightened up the text this paragraph so that it complements rather than repeats the earlier passage, and we think it now more directly motivates the analyses. The broader motivation for climate-demography relationships is addressed by the new Introduction paragraph. [Lines 243-253]

Line 221: [...] I suggest moving Analysis 3 to the supplemental robustness check, and instead focusing on other aspects that influence sample size and uncertainty.

Response: We agree. The former Analysis 3 (the multi-site Astragalus comparison) has been moved to the Supplementary Information and reframed as a robustness check on Analysis 2 (Supplementary Methods S1.10 and Supplementary Results S2.5). The new Analysis 3 is a simulation study with known truth that evaluates how bias and 95% interval coverage vary with effect size and sample size (Figure 6). [Lines 115-116, 409-437]

Line 232: This is a nice summary of these life history metrics, but doesn't give a good sense of why it is biologically interesting that there is a relationship between shape and longevity. [...]

Response: We have expanded the motivation for Analysis 1 to explain why the association between the pace (life expectancy) and shape of mortality is of long-standing interest in the theory of senescence, and why establishing whether it is robust to sampling uncertainty matters for that literature. [Lines 257-266]

Line 308: It is not clear what is meant by matched climate covariates if the authors need to use an external dataset (PRISM) for the climate analysis.

Response: We have clarified that the climate covariates were matched both spatially and temporally: "... matched local climate covariates (PRISM data extracted at the geographic coordinates of the study population and matched in time to the demographic observations)..." [Lines 343-344]

Line 314: It is confusing that the authors state that the only focus on reproduction, when they also analyze survival.

Response: We have corrected the conflicting statement. The text no longer says we focus 'only' on reproduction. It now states that we focus on the reproductive transition (recruitment) and survival. [Lines 350-352]

Line 319: This sentence here is a bit odd in this location in that it references the results that haven't yet appeared [...].

Response: We have reworded this observation so that it no longer pre-empts later results! It is now phrased like this: "Any robustness in this analysis would therefore reflect..." and reads more appropriately as motivation rather than a stated finding. [Lines 355-358]

Line 350: [...] it could be more clear that they are estimating statistical models for both cases, and what is different is that there is some type of measurement error model that accounts for variation in that metric in the second case.

Response: We have added explicit signposting that both formulations are fitted as statistical models, and that the difference is a measurement-error/count-based representation in the second case that propagates sampling variation in the vital-rate estimate. [Lines 376-385]

Line 364: [...] It would be helpful to readers to know what about these methods is actually doing the uncertainty propagation.

Response: We have now clarified what each formulation propagates: we now distinguish an SSD-weighted quasibinomial GLM on point-estimate survival from a binomial count model on $n_{\text{success}}/n_{\text{fail}}$, that propagates sampling uncertainty in stage-specific survival. We also clarify that the weighting and dispersion assumptions differ, so it follows that interval-width differences cannot necessarily be attributed only to sampling uncertainty. [Lines 398-405 and Supplementary Methods S1.6/S1.10]

Line 372: [...] I still think this could be combined with Analysis 2 or moved to the supplement as the results are essentially identical.

Response: Agreed. This analysis (Astragalus) is now in the Supplementary Information as a robustness check on Analysis 2. [Supplementary Methods S1.10 and the Supplementary Results S2.5]

Results

Line 391: I suggest focusing first on the comparison across all the metrics that vary across the spectrum [...], and then move onto these “parameters of interest”. Also noting that here, the authors use parameter instead of trait or metric.

Response: We have added a brief orienting sentence at the start of the Analysis 1 Results that introduces the comparison of point-estimate and posterior-mean variance before the specific regression. After careful consideration, we have kept the current overall order, because we feel that this comparison is most interpretable once the single-MPM example and the focal comparative analysis have been introduced to the reader. We have also standardised our terminology (trait/metric) where it was previously inconsistent. [Lines 440-442]

Line 403: Authors state that the uncertainty only moderately influenced the result, but the actual figure shows a sign flip (although in both cases overlapping zero). So I don't know that it is clearly a moderate influence.

Response: We have revised the description. Rather than calling the effect ‘moderate’, we now state explicitly that the posterior median of the slope changes sign between the point-estimate and sampling-uncertainty models. We do note that this sign change is not trivial, but we also explain that because both posteriors span zero the overall conclusion (that there’s no strong effect) is unchanged. [Lines 457-462]

Line 409: I believe the authors explanation here about posterior widths, but it does not come across clearly in Figure 4 that the posterior uncertainty of the relationship is meaningfully different.

Response: The Analysis 1 Results section now states explicitly that posterior uncertainty in the slope is lower in the sampling-uncertainty model. Figure 4B shows the two posterior densities, and Table S3 gives the numerical summaries. Both intervals include zero, so the narrower interval does not establish a clearly positive relationship. [Lines 463-469]

Line 426: recommend saying “the posterior interval was 10.9% wider”. And also it was not clear which Beta estimate is referred to here. Figure 5B? [...]

Response: We have clarified that the coefficient referred to is the spring-temperature recruitment coefficient (Figure 5B), and now report the difference explicitly as a percentage: the coefficient was 8.0% higher and its 95% interval 10.9% wider in the model incorporating sampling uncertainty. We have distinguished this from the stage-specific survival comparison (Figure 5C) so the two are no longer confusingly conflated. [Lines 479-484]

Discussion

Line 453: it is not clear what is meant by trait dependent. In this case, I would also start the discussion focused on first the results of analysis 1, and then move to analysis 2 later. [...]

Response: In the Discussion, we now clarify trait dependence by distinguishing the sensitivity of mortality-trajectory traits from that of integrated demographic metrics. We keep a brief cross-analysis overview at the outset, beginning with Analysis 1 and then summarising Analysis 2 and the simulation, rather than separating the analyses entirely. Our subsequent discussion explains the contrast among the demographic quantities. [Lines 517-530]

Line 495: I really appreciated this takeaways section. I did wonder how longevity analysis of age-based models would differ. And from analysis that are built directly from the data. [...]

Response: We have now added a short section to the Discussion noting how results might differ for age-based (Leslie) models and for analyses built directly from individual-level data (e.g. IPMs and capture-recapture survival models), where information sharing across ages or stages can alter the magnitude and correlation structure of propagated uncertainty [Lines 595-606].

Figures

Across figures, the points [...] and the sampling uncertainty distributions all appeared in green, but the captions often mentioned purple. [...] This was particularly relevant in Figure 3 and 4 [...].

Response: Thanks for pointing this out. We have now ensured that captions match the plotted colours/symbols. In the two-model comparisons, purple denotes the point-estimate model while green represents the model with sampling uncertainty.

Figures 1 and 2 instead use pale green distributions and teal dashed point-estimate lines. Their captions now describe this correctly. We also removed the redundant legends in Figure 4 to improve readability.

Figure 4: Please state what the main takeaway from the figure is in captions.

Response: We have revised the caption for Figure 4 so that the take-away message is at the beginning: “Accounting for sampling uncertainty shifts the estimated life expectancy-shape relationship and narrows its posterior interval, although both models retain substantial uncertainty about the direction of the relationship.”

Figure S1 and S2 seem like they are showing the same thing, and I think S2 is actually more useful. Also I'm curious if these boundary effects go away if estimating more continuous stages [...].

Response: We have now differentiated the two supplementary figures so their distinct roles are clear (S1 as the boundary-estimate diagnostic, and with S2 as the survivorship-trajectory illustration), and clarified in the Supplementary Results that information sharing in fitted vital-rate models may reduce boundary estimates, whereas finer discretisation alone does not necessarily do so [Figures S1 and S2 and Supplementary Results S2.2]